

Introduction

- 1 - Why do we need statistics?
- 2 - Descriptive statistics
- 3 - Inferential statistics: Principle of statistical test
- 4 - Statistical models and study design
- 5 - Measurements and scales

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1

1

Fred L. Bookstein (2009):

"The success of statistical methods comes at the expense of all the theories that we simultaneously hold to be true about the biological materials to which they both pertain."

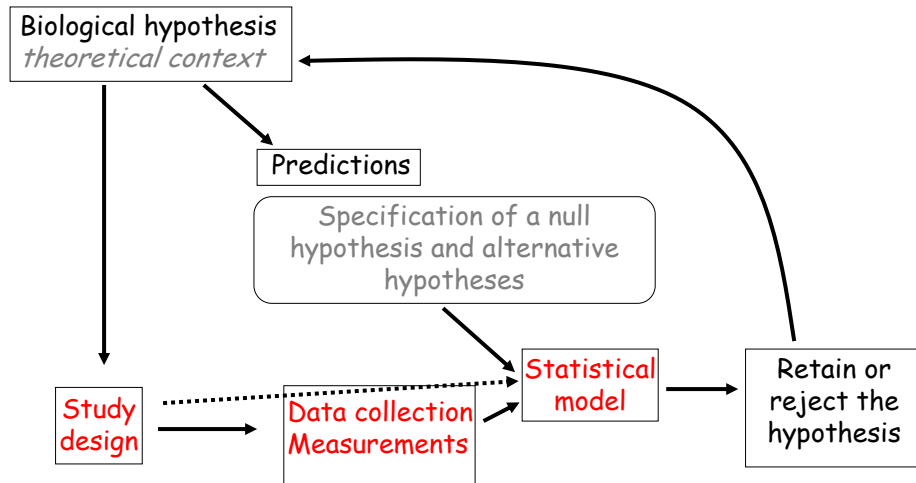
Don't forget about biology

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2

2

Why (and when) do we need statistics?



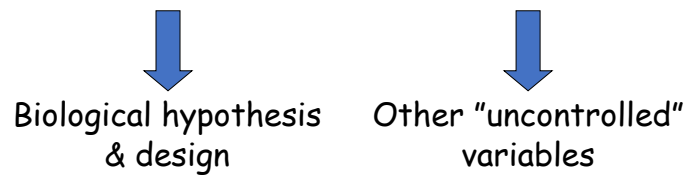
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3

3

Why (and when) do we need statistics?

Observed data = **Structured** part and "Noise"/random variation



$$\text{Response} = \beta_1 \text{ Predictor} + \varepsilon (\sim N(0, \sigma^2))$$

Effect size
(Is the biological effect important?)

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4

4

Why (and when) do we need statistics?

Descriptive statistics

- Statistics used to describe the data: central tendency, variation, skewness, kurtosis

Inferential statistics

- Statistics to make inference about phenomena → testing hypotheses

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5

5

2 - Descriptive statistics:

Distribution

Central tendency

Mean / mode / median

Dispersion

Variance / standard deviation

Standard error of the mean / confidence interval

Shape parameters

Skew

Kurtosis

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6

6

3 - Inferential statistics: Basic principles of a significance test

We take the "null" hypothesis* as a reference

Find evidence against the null hypothesis when we observe a sample which is unexpected under this null hypothesis

1. Need to describe the null hypothesis in terms of a statistical model that can be used to derive the sampling distribution of sample statistics (i.e. to measure if something is expected or not),
2. We only measure the evidence against the null hypothesis. Therefore, this cannot be used to say if evidence is supportive (never confirmatory)

* The null hypothesis designate the simplest model, that is, the model with the lowest number of parameters

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7

7

Basic principle of a significance test: the t-test

The hypothesis is defined in terms of parameters

In the t-test: difference between two means = β
(β is an unknown parameter)

1. $H_0 : \beta = 0$
2. Test Statistic $\rightarrow \beta = \bar{x}_1 - \bar{x}_2$
3. Larger value of $\beta \rightarrow$ Stronger evidence against the null hypothesis (H_0 is less likely)

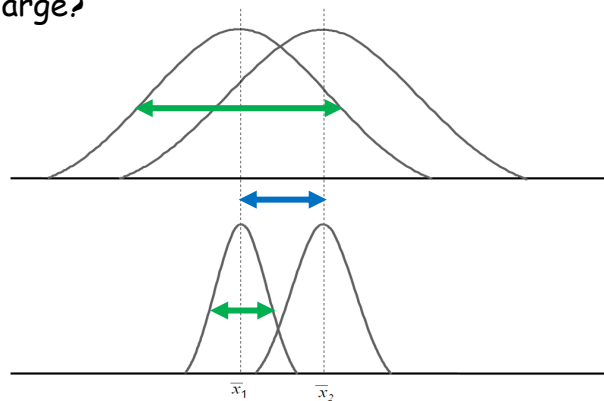
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8

8

Basic principle of a significance test: the t-test

When is β large?



Difference between groups = **structured part** (biological hypothesis)
 Difference within groups = **random variation** (residuals)

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9

9

Basic principle of a significance test: the t-test

A statistical model compares the proportion of variance of the response variable due to the **predictor variable(s)** with the proportion of variance due to **random variation**.

Larger is the former relative to the latter, stronger is the evidence against H_0 .

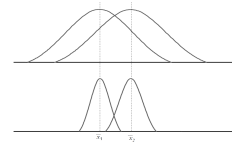
$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \longrightarrow \beta$$

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10

10

Basic principle of a significance test: the t-test



The T statistic will compare the relative importance of the structured and random parts of the variation

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Between treatments variation
= variation due to biological factor

Within treatment variation
= variation due to «noise»

When is T large, or large enough to falsify H_0 ?

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11

11

Basic principle of a significance test: the t-test

Sampling distribution: possible values of T when H_0 is true

Finding the sampling distribution of the T statistic with



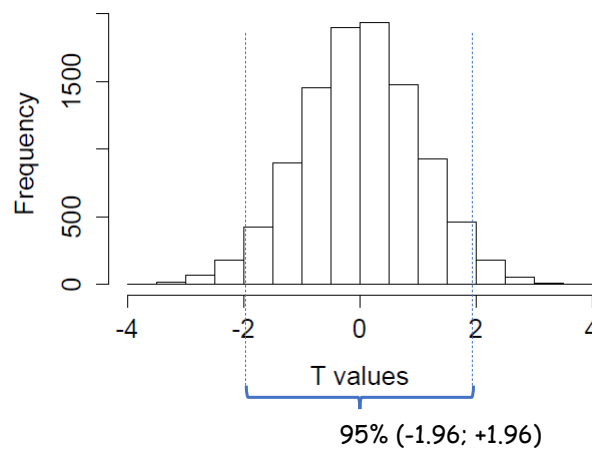
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12

12

Basic principle of a significance test: the t-test

Sampling distribution of T under H_0



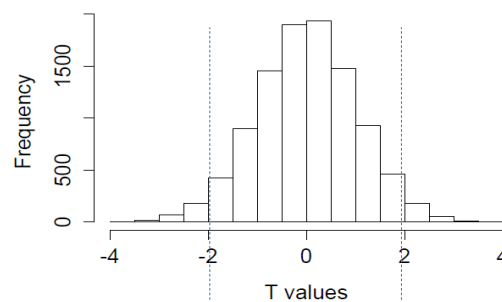
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13

13

Basic principle of a significance test: the t-test

Probability (P) value



$$P\text{-value} = \text{Prob} (T(X) \geq T(X_{\text{obs}}) \mid H_0)$$

P value = probability of observing samples as extreme or more extreme than the one observed, given H_0 is true.
(extreme defined according to α)

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14

14

Basic principle of a significance test: the t-test

Type I and type II error and statistical power

Common practice: reject H_0 if P-value < 0.05

"not reject"/ "accept" H_0 if P-value > 0.05

	H_0 Accepted	H_0 Rejected
H_0 True	correct	Error I α
H_1 True	Error II β	correct

α = probability that the test leads to a type I error

β = probability that the test leads to a type II error (1-power)

Power = 1 - β

15

15

Basic principle of a significance test: the t-test

Power of the test = probability of not making a type II error

probability that an experiment will detect an effect as significant when the effect exists.

What will affect the power of detecting a biological effect?

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

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16

16

Statistical power

The power of a statistical test depends among other things on the number of independent observations.

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

→ Effect size / biological effect
 → Sample variance: "random" variance + measurement error
 → Sample size: number of independent observations

The number of observations and the measurement error depend on the study design

17

Interpreting the results: statistical vs. biological significance

```
> ss1<-rnorm(50, 10,1)#random sampling of 30 individuals in a population
with mean=0, SD=1
> ss2<-rnorm(50, 11,1)#random sampling of 30 individuals in a population
with mean=1, SD=1
> t.test(ss1,ss2)
```

Welch Two Sample t-test

```
data: ss1 and ss2
t = -4.3495, df = 97.952, p-value = 3.347e-05 ← Statistical significance
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-1.1798990 -0.4405535 ← 95% CI of the difference between the two populations
sample estimates:
mean of x mean of y
0.1911812 1.0014074 ← means
```

Biological significance: 0.19 - 1.00 = -0.81 difference between the two populations

18

Statistical Significance ≠ Biological Significance

Pay attention to the effect size and their precision!!!

To do that:

Always write the units after the parameter estimates

Ask yourself the questions:

What does these numbers mean?

Is it biologically relevant?

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19

19

4 - Statistical models

Observed data = Structured part and "Noise"/random variation

$$y_i = a_i x_i + \varepsilon_i$$

Response variable Predictor variable Residual variance

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20

20

Statistical models

Observed data = Structured part and "Noise"/random variation

$$y_i = ax_i + \varepsilon_i$$

The model will be described in terms of parameters: a

The number of parameters depends on the structure of the model, that is, the type (discrete vs. continuous) and the number of predictor variables.

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21

21

Statistical models and experimental design

Model complexity

The complexity of the statistical model and the number of parameters to estimate depend on the number of predictor variables, their type, and their possible interactions, and on the structure of the residual variation.

In turn, the number of predictor variables, their type, and possible interactions, as well as the structure of the residual variation, depend on the design of the study.

22

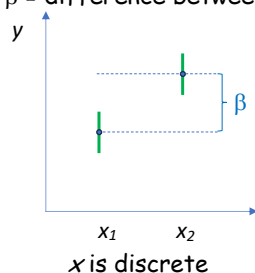
Statistical models

Observed data = Structured part and "Noise"/random variation

$$y_{ik} = \alpha_i x_{ik} + \varepsilon_{ik}$$

$$H_0: \beta = 0, H_1: \beta \neq 0$$

where β = difference between the two means

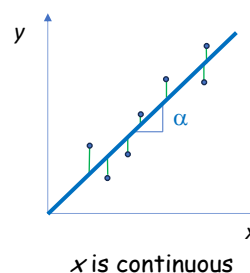


Model parameters

two means μ_1 & μ_2

$$y_i = \beta + \alpha x_i + \varepsilon_i$$

$$H_0: \alpha = 0, H_1: \alpha \neq 0$$



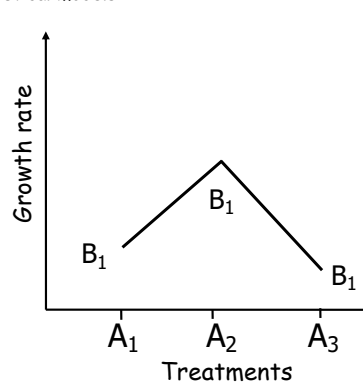
1 slope α , 1 intercept β

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23

23

Statistical models

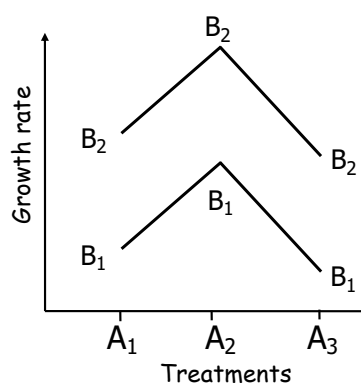


Model

$$y_{ik} = \mu + \alpha_i + \varepsilon_{ik}$$

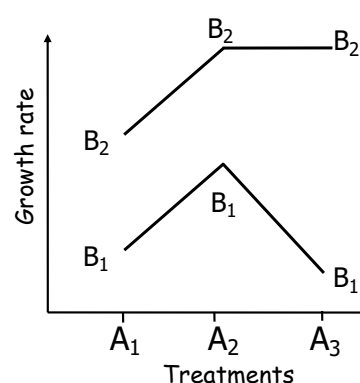
Number of parameters

3
diff between 3 treatments



$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}$$

4
Diff between 3 treatments,
diff between 2 genotypes

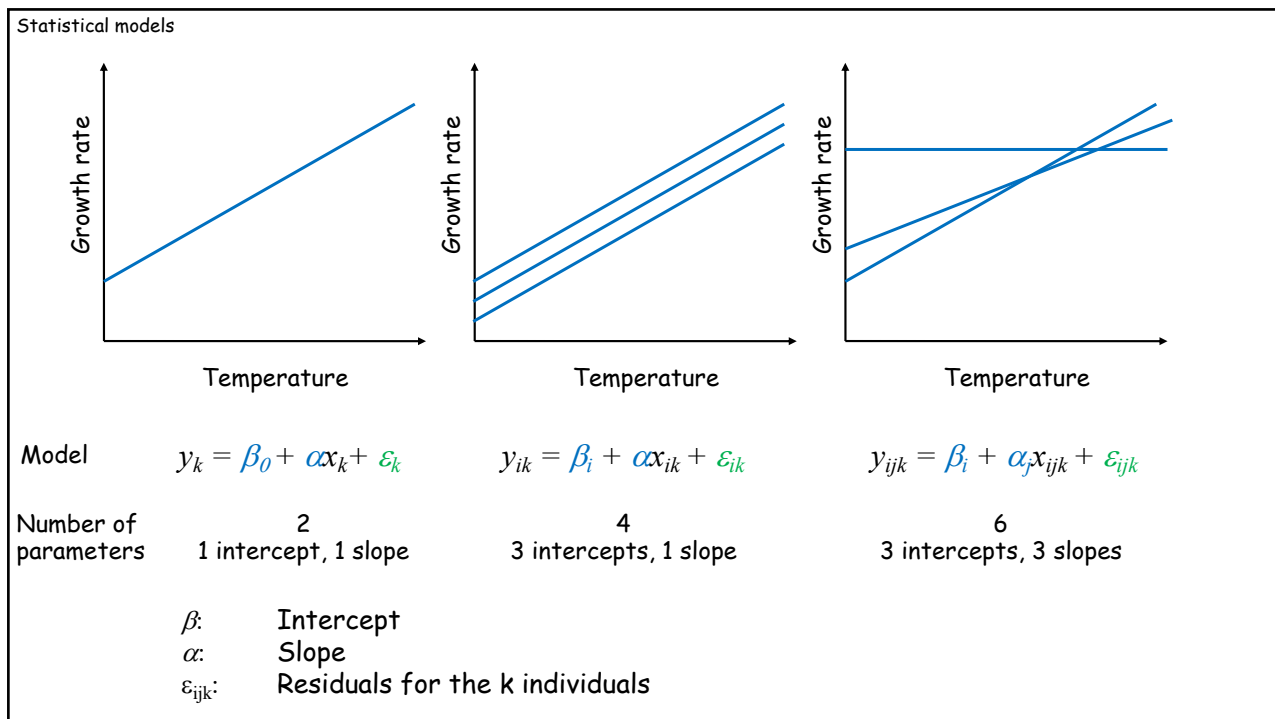


$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

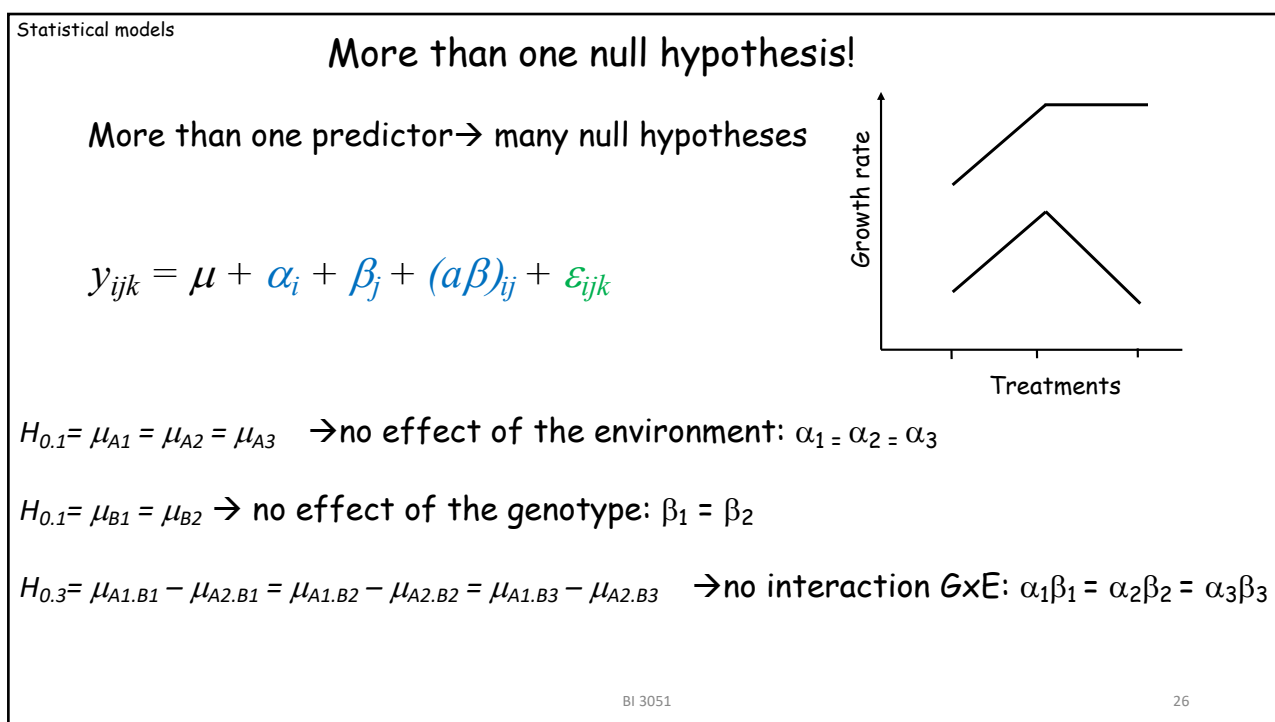
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Diff between 3 treatments
for the 2 genotypes

μ : Grand mean
 α_i, β_j : Parameters describing the effects of the predictor variables
 ε_{ijk} : Residuals for the k individuals

24



25



26

Linear models: definition

A linear model refers to a statistical model where the response variable is described by a linear combination of parameters, and no parameters appears as an exponent or is multiplied or divided by another parameter.

linear model $y = \beta + \alpha x + \gamma z$

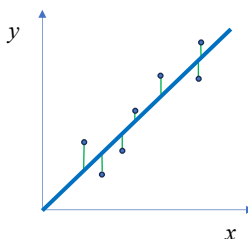
nonlinear model $y = \alpha x^\beta$

27

Sample size, and the number of independent observations

An important assumption of the linear models: residuals are independent.

$$y_i = ax_i + \varepsilon_i$$

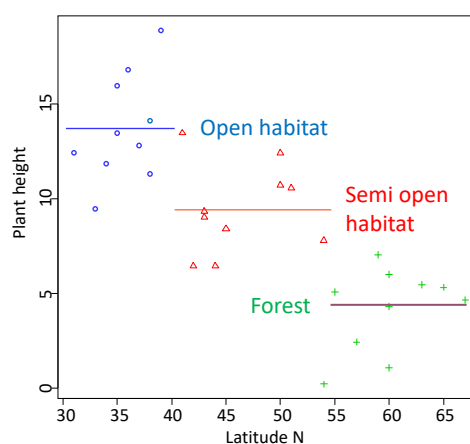
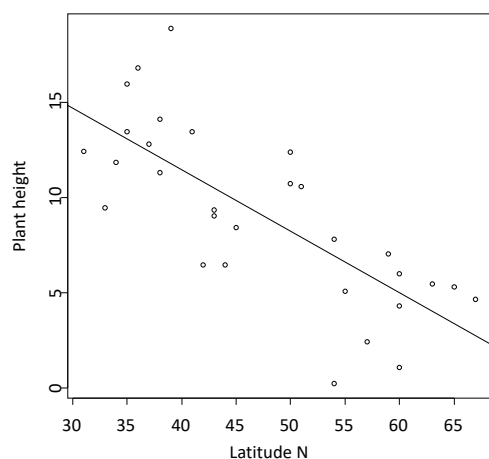


Non-independent residuals will affect the magnitude of residual variance (increase or decrease it) and consequently the result of the statistical test.

When designing a study / experiment, we must consider the factors that can affect the independence of the observation and consequently the independence of the residuals.

28

Confounding factor(s)



Habitat is a confounding factor that generates non-independence among observations

29

Pseudoreplication

Testing the effect of a chemical: three concentrations + control

Four growth chambers / incubators

How do you do?



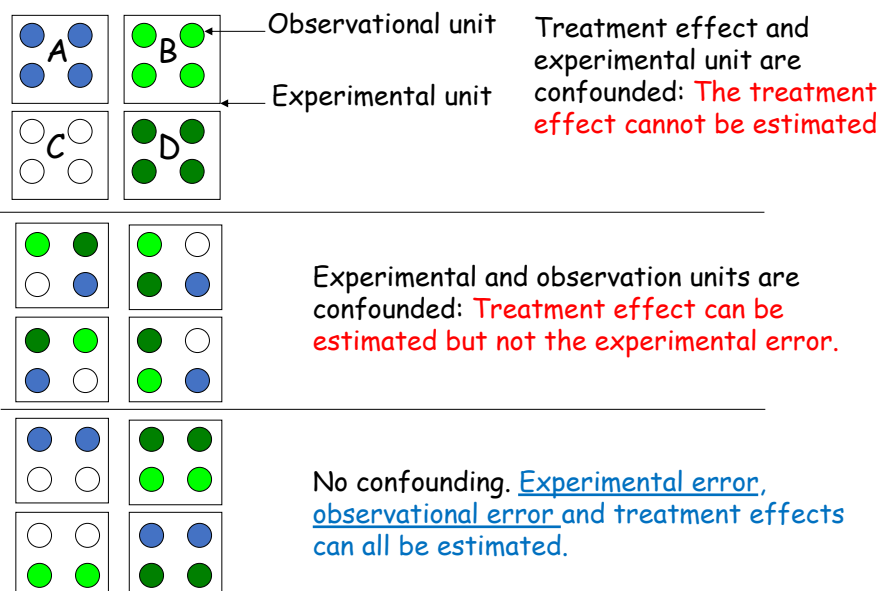
Observational units
(3 + 1 treatments)



Experimental unit
(growth chamber)

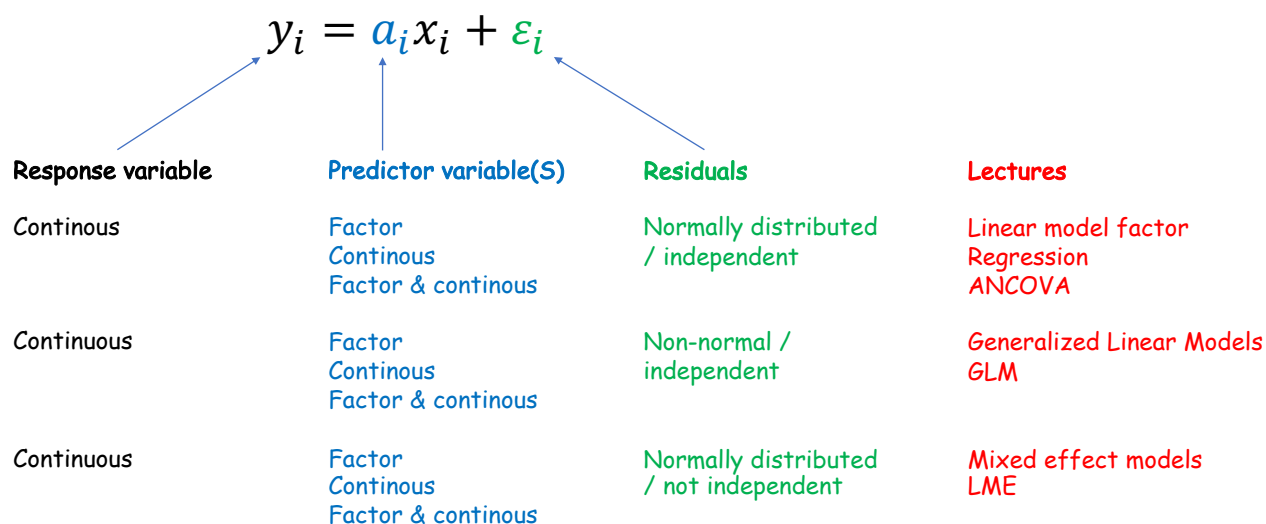
30

Statistical models and study design



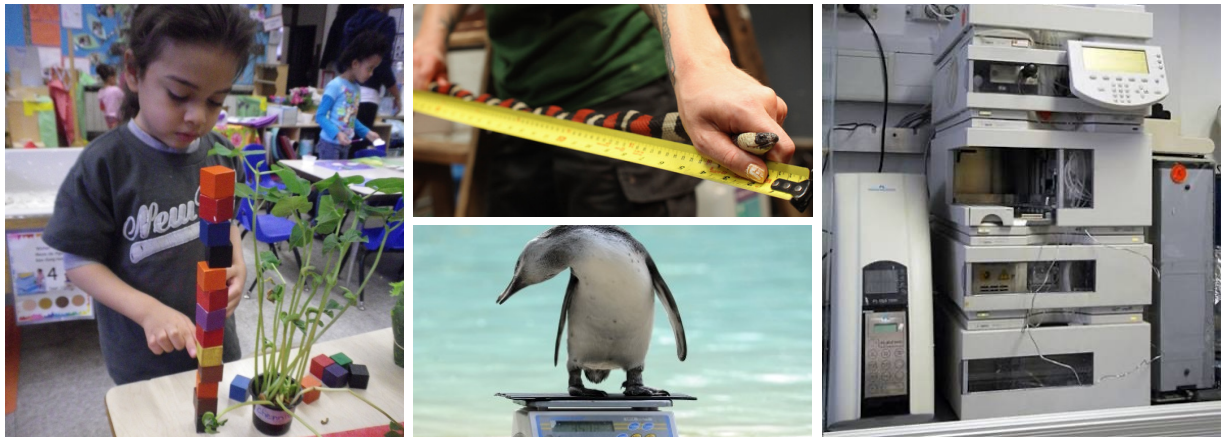
31

Statistical models and study design



32

5 - Measurements and scales



33

Meaning of measurement

Measurement consists of an assignment of numbers to attributes of entities so that the relations between the numbers can capture empirical relations among the attributes

Houle et al. 2011

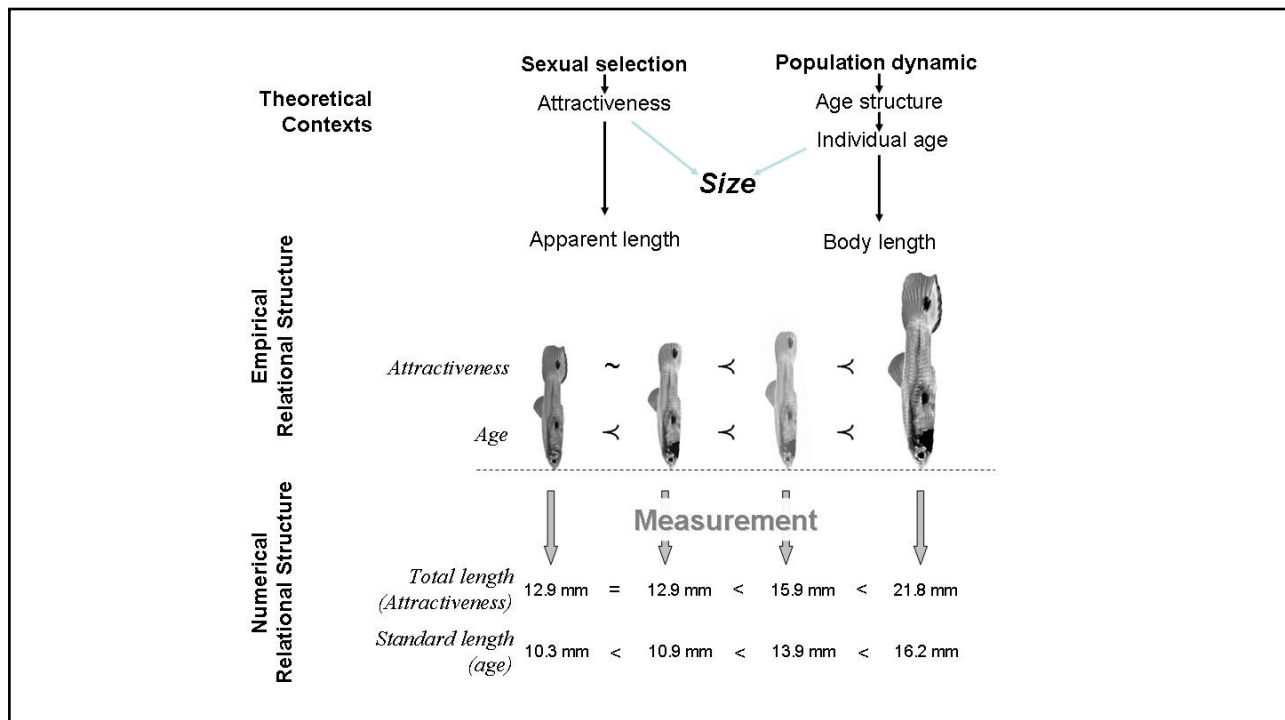
34

Am I measuring the right things?



How to measure size? → Why?

35



36

Meaning of measurement

A quantity consists of a name, symbol, procedural statement, numbers and unit of measurement

Schneider 2009

Any numbers that refer to a quantity must always be associated with its unit!

37

Scales of measurements

The scale types refer to mappings that preserve the relations between attribute in the measurements

The mathematical rules that apply to units on a particular scale differ from those that apply to numbers

38

Scales of measurement

Scale	Domain	meaningful comparisons	Examples
Nominal	Any set of symbols	equivalence	Species, genes
Ordinal	Ordered symbols	order	Dominance, preference
Interval	Real numbers	order, differences	Dates, temperatures ($^{\circ}\text{C}$), Malthusian fitness
Ratio	Positive real numbers	order, ratios, differences	Length, mass, duration
Signed ratio*	Real numbers	order, ratios, differences	signed asymmetry, intrinsic growth rate (r)
Absolute		any	probability

39

Scales of measurement Permissible transformation

Mathematical rules that apply to units define the meaningful (permissible) transformations that can be applied to the numbers

Permissible transformations are transformation that don't change the properties of the number for a given scale

40

Permissible vs. non-permissible transformation

Properties of the numbers

$$a/b > c/d \text{ and } a - b > c - d \quad a = 3; b = 5; c = 4; d = 7$$

$$3/5 = 0.6 > 4/7 = 0.571 \quad a/b > c/d \text{ True}$$

$$3 - 5 = -2 > 4 - 7 = -3 \quad a - b > c - d \text{ True}$$

Multiplication by a constant: 2

$$6/10 = 0.6 > 8/14 = 0.571 \quad a/b > c/d \text{ True}$$

$$6 - 10 = -4 > 8 - 14 = -6 \quad a - b > c - d \text{ True}$$

Log-transformation

$$\log(3)/\log(5) = 0.683 < \log(4)/\log(7) = 0.712$$

$$a/b > c/d \text{ False}$$

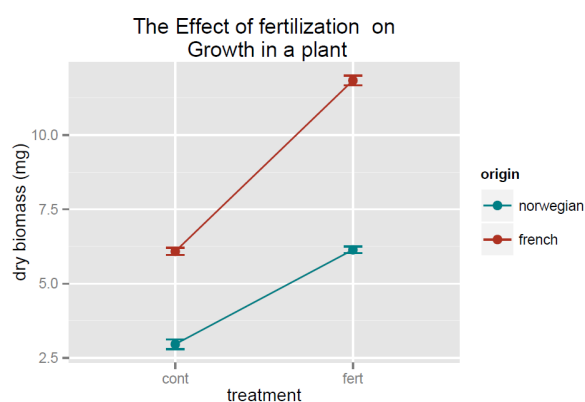
$$\log(3) - \log(5) = -0.511 > \log(4) - \log(7) = -0.559$$

$$a - b > c - d \text{ True}$$

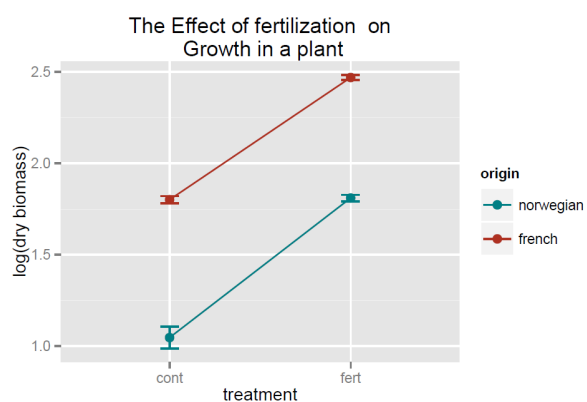
41

Transformation and biological meaning

Raw data:
genotype x environment interaction



Log-transformed data:
no genotype x environment interaction



42